

Detecting Stabbing Motions Using Smartwatch Sensor Data: A Forensic Machine Learning Proof-of-Concept

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Abstract—Smartwatches continuously record detailed motion and physiological data, making them a potentially valuable source of digital trace evidence in forensic investigations. This study provides a proof-of-concept for the use of smartwatch inertial sensor data to distinguish a simulated stabbing motion from operationally similar everyday activities. Data were collected from three participants using a Samsung Galaxy Watch 7 and the SensorLogger application, capturing accelerometer, gyroscope, gravity, orientation, and total acceleration data across five activity classes: stabbing, vacuuming, brushing teeth, brushing hair, and a resting baseline. Three tree-based ensemble classifiers – Random Forest, CatBoost, and XGBoost – were trained and evaluated using a random 80/20 split and a leave-one-participant-out (LOPO) strategy. Under the random split, all models achieved approximately 99% accuracy with perfect stabbing recall. The LOPO evaluation revealed an asymmetry: when trained on two participants and tested on a third, stabbing recall remained high at 0.984; when trained on a single participant and tested on the remaining two, recall collapsed to between 0.107 and 0.232. This asymmetry indicates that stabbing motion exhibits meaningful inter-individual variability, and that diverse multi-participant training data are necessary for reliable generalisation. These findings suggest that smartwatch sensor data holds forensic potential for activity classification, but that substantial further validation with larger and more diverse datasets is required before operational deployment can be considered.

Index Terms—smartwatch forensics, human activity recognition, stabbing detection, machine learning, leave-one-participant-out evaluation

I. INTRODUCTION

Wearable smart devices are becoming increasingly common in everyday life. Fitness trackers and smartwatches in particular have shown rapid expansions in the global technology market, with more growth expected in the coming years [1], [2]. This growth is coupled with an increased interest in health monitoring, general digital lifestyles, communication, and activity tracking [3]. These devices are typically equipped with multiple sensors that continuously record physiological and spatial data, generally stored locally on the device or a paired smartphone for short-term use and long-term through online, often cloud-based, storage services. Access to this data is often restricted due to data protection regulations, such as

the General Data Protection Regulation (GDPR) [4], but it still exists and can be accessed with the proper permissions and legal justification.

Many smartwatches are capable of measuring heart rate through photoplethysmography [5] and movement parameters such as linear acceleration through accelerometers and rotation through gyroscopes, with precise timestamps. These sensors have been validated for measuring movement and orientation during physical activity [6]. Together, they allow smartwatches to provide a comprehensive summary of the movements made by the wrist and the physiological state of the wearer during a given time window. Combined with the long-term storage of this data, this creates the potential for inferences to be made as to the activity of the user in a forensic context. In particular, competing propositions from the prosecution and defense can be assessed for compatibility with smartwatch data at a specific time of interest, in order to garner support for one proposition over the other.

Recent studies have explored whether this kind of data can be used in a forensic context, particularly for activity classification. Most of this research has focused on smartphone data and distinguishing different movement patterns using machine learning [7]–[12]. In contrast, the use of smartwatch data in forensic contexts is still in an early phase, with some research pointing to its potential [13] but limited validation studies available [14], [15]. Nevertheless, the large body of research on smartphone data, which uses similar sensors and produces similar outputs, suggests that smartwatches may hold comparable forensic potential. If validated, this could provide an invaluable and continuously growing source of digital evidence for scenario reconstruction of movements or actions at specific points in time.

The extent to which smartwatch sensor data can reliably differentiate between operationally similar wrist or arm movements is currently understudied and not yet ready for forensic application [13]. Many everyday activities involve repetitive movements that are operationally similar to motions associated with (potential) criminal acts. Thus, these could easily be confused without the proper means to differentiate them.

To understand the potential evidentiary value of smartwatch sensor data, it must first be explored whether such activities can be distinguished, and if so, to what degree.

The present study provides a proof-of-concept for the use of smartwatch sensor data in distinguishing between specific wrist and arm movements – specifically, whether a simulated stabbing motion can be differentiated from operationally similar everyday activities including vacuum cleaning, brushing hair and teeth, and a resting baseline. Three tree-based ensemble classifiers were applied to features extracted from five sensor streams recorded using a Samsung Watch 7. Both a standard random split and a leave-one-participant-out evaluation were employed to assess generalisation to unseen individuals, which is particularly relevant to forensic application. The central research question is: *can accelerometer, gyroscope, gravity, orientation, and total acceleration data from a wrist-worn smartwatch distinguish a simulated stabbing motion from operationally similar everyday wrist and arm movements?*

II. METHODOLOGY

A. Data Collection

Sensor data were recorded using a Samsung Watch 7 worn on the dominant (right) hand of each participant. Two applications were used in parallel: the SensorLogger application (version 1.52.0) [16], which provided access to raw sensor streams at high temporal resolution, and the Samsung Health native application, which was initially used to record photoplethysmographic [17] heart rate data. Five sensor streams were captured: three-axis accelerometer, three-axis gyroscope, three-axis gravity, orientation (expressed as both Euler angles and quaternion components), and three-axis total acceleration. Data were exported as comma-separated value files following each recording session. All raw sensor recordings, annotation files, and analysis code are publicly available at GitHub repository.

Three participants took part in data collection, all members of the research team. Participants 1 (Felicie) and 2 (Mara and Lotte, treated as a combined group; see Section II-C) performed five activity classes: *vacuuming*, *brushing_teeth*, *brushing_hair*, *doing_nothing*, and *stabbing*. The stabbing motion was performed in two variants – overhead and forward thrust – under both rested and physically elevated heart rate conditions, the latter induced through brief aerobic exercise prior to recording. Activity onset and offset times were manually logged by one of the two uninvolved participants during recording using an external clock, and subsequently used to label the corresponding sensor data.

Each activity was performed for two minutes per session, with a minimum rest interval of 90 seconds between activities to allow heart rate and physiological conditions to return to baseline. The stabbing motion was performed against a hard surface — specifically the back of an office chair positioned to approximate the torso height of a standing adult — covered with layers of clothing to protect participants’ hands. This setup was chosen to provide realistic physical resistance while

ensuring participant safety. For the elevated heart rate condition, participants performed jumping jacks for one minute prior to each activity, followed by a 30-second stabilisation period, before beginning the two-minute recording window. All rested condition recordings were completed before any elevated condition recordings within each session, to prevent residual heart rate elevation from contaminating the rested baseline.

The selection of daily activities was motivated by their operational similarity to the stabbing motion, particularly with respect to repetitive arm and wrist kinematics. Vacuuming, hair brushing, and teeth brushing all involve cyclical wrist movements that could plausibly confound a classification system, making them appropriate negative controls for evaluating the discriminability of the stabbing motion.

Forensic context and tool selection.

In operational forensic casework, raw sensor streams of the kind captured by SensorLogger would not typically be accessible: offenders are unlikely to have third-party logging applications installed on their devices, and standard digital forensic extraction tools such as Cellebrite [18] operate on data stored by manufacturer applications. Samsung Health data can be requested via a General Data Protection Regulation (GDPR) subject access request directed to Samsung, or extracted from a seized device via Cellebrite. In the latter, the resulting data provides only aggregate activity summaries rather than individual raw sensor readings required to distinguish specific wrist movements. Additionally, the GDPR data request pathway was not pursued in the present study due to time constraints. SensorLogger was therefore adopted as a research instrument under the explicit acknowledgement that the present study constitutes a controlled proof-of-concept rather than a pipeline directly deployable in casework. Future work may explore whether partial raw signal recovery from manufacturer-stored data could bridge this gap.

B. Data Quality and Recording Gaps

Data collection was affected by intermittent recording gaps arising from the interaction between the SensorLogger application and the Android operating system of the Samsung Watch 7. The Samsung OS aggressively suspends background processes to conserve battery life, causing SensorLogger to cease writing data mid-session without user notification. The presence and OS-level origin of these gaps was confirmed by inspecting the `seconds_elapsed` column alongside wall-clock timestamps: in all cases, both values exhibit simultaneous jumps of identical magnitude, indicating that no data was written during the missing intervals and that the absent observations cannot be recovered post hoc.

Critically, recording gaps were observed during the *doing_nothing* baseline condition, in which participants remained completely still. This confirms that the gaps are attributable to OS-level process suspension rather than to physical movement causing the watch display to wake and interrupt recording. The cause was independently confirmed by the developer of SensorLogger (K. Sheridan, personal communication, March

2026), who identified Bluetooth connection loss between the watch and the paired smartphone as the mechanism responsible for data loss on Android devices. The developer noted that a hardware-side buffering solution, in which data are retained on the watch and transferred at the end of a session, is under active development. Because the gaps arise from an OS- and connectivity-level mechanism unrelated to the activity being performed, they are not expected to introduce systematic bias toward or against any particular activity class.

Heart rate recording was discontinued following the observation that the Samsung Health native exercise application – which was required to obtain heart rate measurements at a sampling rate of approximately 1 Hz, as opposed to the once-per-ten-minutes passive monitoring rate of the standard Samsung Health application – introduced additional recording gaps when operated simultaneously with SensorLogger, beyond those attributable to OS suspension alone. Heart rate data were therefore excluded from all subsequent analysis.

Two sessions were lost in their entirety: the stabbing (elevated heart rate) session of Participant 1, and the vacuuming (rested heart rate) session of Participant 2. Both were re-recorded on multiple occasions; however, the recovered datasets remained too sparse to contribute meaningfully to analysis and were excluded. As a consequence of the latter loss, the vacuuming class contains data from the elevated heart rate condition only. This is noted as a limitation but does not affect the analysis, as the rested/elevated distinction was removed during label design (Section II-C).

C. Preprocessing and Label Design

Heart rate and condition labels. The rested and elevated heart rate conditions were collapsed into a single label per activity. Since heart rate data were excluded and the two conditions involve physically identical movements, retaining the distinction would have introduced label granularity without analytical value.

Stabbing label consolidation. The overhead and forward stabbing variants were merged into a single *stabbing* class. In forensic casework the direction of a stabbing motion is typically unknown at the point of analysis, and both variants may occur within a single incident. Merging the two variants improves class coverage across participants and better reflects the operational scenario the classifier is intended to address.

Participant structure. Participants Mara and Lotte were combined into a single participant group (Participant 2). Both individuals performed all assigned activities, and their recordings provide complementary coverage of the activity classes. This grouping increases the volume of training data available for the leave-one-participant-out evaluation described in Section II-F.

Sparse row removal. Five rows corresponding to annotation segments containing fewer samples than a single analysis window were removed, as they could not contribute to feature extraction.

The resulting dataset comprised 59,213 rows across five activity classes, distributed as shown in Table I.

TABLE I
LABEL DISTRIBUTION IN THE FINAL DATASET.

Label	P1 (Faye)	P2 (Mara+Lotte)	Total
vacuuming	370	1,053	1,423
brushing_teeth	2,173	1,826	3,999
brushing_hair	19,292	4,707	23,999
doing_nothing	9,598	4,331	13,929
stabbing	4,611	11,252	15,863
Total	36,044	23,168	59,213

D. Downsampling

Raw sensor data were recorded at an irregular sampling rate, varying between approximately 50 and 100 Hz depending on the session and the specific sensor stream. To ensure that all analysis windows contain an identical number of rows – a requirement for consistent feature extraction – the data were resampled to a uniform rate of 30 Hz using time-based linear interpolation. Resampling was performed independently on each contiguous, single-label segment (defined by a continuous block of rows sharing the same activity label and performed by the same participant). This segmentation strategy ensures that interpolation is never applied across annotation boundaries or recording gaps: where a gap exists, the resampled time grid will contain no data, preserving the integrity of the gap rather than fabricating values across it. Following resampling, the dataset contained 37,108 rows.

E. Sliding Window Feature Extraction

The resampled time series were segmented into overlapping windows of fixed duration, following the sliding window paradigm commonly used in human activity recognition research [19]. Each window spans two seconds, corresponding to 60 rows at 30 Hz. A window duration of two seconds was selected to capture a complete stabbing motion, including both the forward thrust and the return phase. Previous biomechanical studies report that individual knife motions typically occur within approximately 0.62–1.07 seconds [20]. A two-second window is therefore sufficient to capture the full motion while remaining short enough to avoid blending consecutive activities.

Consecutive windows were advanced by a one-second step (50% overlap), doubling the effective window count and reducing the likelihood that a motion occurring near a window boundary is missed. As is common in the human activity recognition literature [19], overlapping windows introduce correlation between adjacent segments – window N and window $N + 1$ share 30 rows. In the random 80/20 split, this means training and test windows are not fully independent, which is likely to inflate performance estimates for that condition. This is acknowledged as a limitation. In the leave-one-participant-out evaluation, no such concern applies: windows are extracted independently per participant, so training and test sets share no data under any overlap configuration.

No window was permitted to cross an activity label or participant boundary. Segments containing fewer than 60 rows following resampling were excluded. For each window, five

summary statistics – mean, standard deviation, minimum, maximum, and range – were computed across each of the 19 sensor columns, yielding a feature vector of 95 values per window. Window counts per activity class are reported in Table II.

TABLE II
NUMBER OF ANALYSIS WINDOWS PER ACTIVITY CLASS.

Label	Windows
brushing_hair	356
brushing_teeth	140
doing_nothing	292
stabbing	351
vacuuming	79
Total	1,218

F. Train/Test Splits

Two complementary evaluation strategies were employed.

Random 80/20 split. Windows were split randomly into a training set (80%, $n = 974$) and a test set (20%, $n = 244$), stratified by class. This provides a baseline estimate of model performance when training and test data are drawn from the same participant population. As noted above, performance under this split is likely to be inflated due to window overlap and should be interpreted as an upper bound rather than an operational estimate.

Leave-one-participant-out (LOPO). Each participant group served as the held-out test set in turn, with the remaining group used for training. This yields two conditions: training on P2 and testing on P1 ($n_{\text{train}} = 681$, $n_{\text{test}} = 537$), and training on P1 and testing on P2 ($n_{\text{train}} = 537$, $n_{\text{test}} = 681$). The LOPO evaluation directly tests generalisation to an unseen individual, analogous to a forensic scenario in which a trained model is applied to data from an unknown suspect. LOPO is considered the primary evaluation for drawing conclusions about forensic applicability.

G. Classification Models

Three tree-based ensemble classifiers were trained and evaluated: Random Forest [21], XGBoost [22], and CatBoost [23], selected on the basis of their suitability for structured feature matrices. All models were trained with 200 estimators and a fixed random seed to ensure reproducibility; no hyperparameter optimisation beyond these defaults was performed.

Random Forest constructs an ensemble of decision trees, each trained on a bootstrap sample of the data and a random feature subset, with predictions determined by majority vote. The method is robust to overfitting and yields interpretable feature importance scores [21].

CatBoost employs ordered boosting to reduce prediction bias and requires minimal hyperparameter tuning relative to comparable gradient boosting implementations [23].

XGBoost builds trees sequentially, with each successive tree trained to correct the residual errors of the current ensemble. The method is computationally efficient and performs consistently well on tabular data [22].

The primary evaluation metric is *stabbing recall*: the proportion of true stabbing windows correctly identified as such. In a forensic context, a false negative – failing to detect a stabbing motion – carries greater consequence than a false positive, and recall is therefore prioritised over overall accuracy or F1-score. Secondary metrics reported include precision, F1-score, overall accuracy, and confusion matrices for all nine model-split combinations.

III. RESULTS

Three ensemble classifiers – Random Forest, XGBoost, and CatBoost – were evaluated under three split conditions: a random 80/20 split and two leave-one-participant-out (LOPO) conditions. The primary evaluation metric is stabbing recall; secondary metrics are overall accuracy, precision, and F1-score. Summary results are presented in Table III. Confusion matrices for all nine model-split combinations are provided in Figures 1 and ??.

TABLE III
MODEL PERFORMANCE ACROSS ALL SPLIT CONDITIONS. STABBING RECALL IS THE PRIMARY FORENSIC EVALUATION METRIC.
P1 = PARTICIPANT 1 (FAYE); P2 = PARTICIPANT 2 (MARA AND LOTTE).

Split	Model	Acc.	Stab. Rec.	Stab. F1
Random 80/20	RF	0.992	1.000	1.000
Random 80/20	XGB	0.992	1.000	1.000
Random 80/20	CB	0.996	1.000	1.000
LOPO (test P1)	RF	0.596	0.984	0.455
LOPO (test P1)	XGB	0.521	0.984	0.450
LOPO (test P1)	CB	0.534	0.984	0.427
LOPO (test P2)	RF	0.443	0.232	0.376
LOPO (test P2)	XGB	0.438	0.215	0.353
LOPO (test P2)	CB	0.388	0.107	0.193

RF = Random Forest; XGB = XGBoost; CB = CatBoost

A. Random 80/20 Split

Under the random 80/20 split, all three models achieved near-perfect performance: overall accuracy ranged from 99.2% (Random Forest and XGBoost) to 99.6% (CatBoost), and stabbing recall was 1.000 across all classifiers. The consistency of these results across three independently trained models indicates that the extracted features carry a strong and learnable signal when training and test data are drawn from the same participant population.

These results should be interpreted with caution. As discussed in Section ??, the 50% window overlap introduces statistical dependence between adjacent windows, such that training and test sets in the random split are not fully independent. Performance under this condition is therefore best interpreted as an upper bound on achievable accuracy under ideal conditions rather than as a reliable estimate of operational performance. The LOPO results presented below constitute the primary basis for evaluating forensic applicability.

B. LOPO Evaluation: Training on P2, Testing on P1

When trained on Participant 2 and evaluated on the unseen data of Participant 1, overall accuracy ranged from 52.1% (XGBoost) to 59.6% (Random Forest). Despite this reduction

in overall performance, stabbing recall was 0.984 across all three models, indicating that 98.4% of true stabbing windows in the Participant 1 test set were correctly identified.

Stabbing F1-score was substantially lower than stabbing recall (0.427-0.455), reflecting reduced precision of approximately 0.27-0.30. The models therefore produced a considerable number of false positive stabbing predictions. In the forensic context of this study, this tradeoff is acceptable: a classifier that reliably detects stabbing motions at the cost of some false alarms is preferable to one that misses genuine events. It should be noted, however, that the high false positive rate would limit the evidentiary value of such a system in formal legal proceedings, where claims must meet a higher standard of specificity. The current results are therefore best suited to an investigative context, where flagging potential matches for further review is the primary goal.

Performance varied across the remaining activity classes. The *doing_nothing* class was classified with reasonable recall (0.75–0.79). The *brushing_teeth* class showed a pronounced collapse in recall (0.06–0.24 across models), suggesting that Participant 1’s teeth-brushing motion profile differs substantially from the training data. The *vacuuming* class was represented by only three windows in the Participant 1 test set, rendering results for that class uninterpretable. The key finding from this condition is that stabbing detection transfers well from a multi-individual training set to an unseen individual, even when overall generalisation across all activity classes is imperfect.

C. LOPO Evaluation: Training on P1, Testing on P2

The reverse LOPO condition – training on Participant 1 alone and testing on Participant 2 – revealed a critical asymmetry relative to the previous condition. Overall accuracy ranged from 38.8% (CatBoost) to 44.3% (Random Forest), and stabbing recall collapsed to between 0.107 (CatBoost) and 0.232 (Random Forest). The majority of true stabbing windows were misclassified, with the *doing_nothing* class achieving perfect recall (1.000) across all three models.

Stabbing precision remained very high (0.97–1.00): on the rare occasions that a stabbing prediction was made, it was almost always correct. The failure was therefore one of sensitivity rather than specificity – the classifiers recognised the stabbing signature when they predicted it, but the training data from Participant 1 alone was insufficient to cover the range of stabbing motion profiles present in the Participant 2 test set. This is consistent with the class distribution: while Participant 1 contributes 351 stabbing training windows, all originate from a single individual, limiting the inter-individual variability captured during training. The Participant 2 test set, by contrast, contains 289 stabbing windows from two individuals with distinct motion profiles.

The *vacuuming* class was entirely undetected (recall 0.000) in this condition, as the Participant 1 training data contained insufficient vacuuming examples – arising from the session loss described in Section II-B – to enable generalisation to Participant 2.

D. Asymmetry Between LOPO Conditions and Key Finding

The pronounced asymmetry between the two LOPO conditions constitutes the primary finding of this study. When models were trained on data from two individuals (Participant 2) and tested on a third unseen individual (Participant 1), stabbing recall was near-perfect at 0.984. When the direction was reversed – training on a single individual and testing on a multi-individual group – stabbing recall fell to between 0.107 and 0.232. This asymmetry demonstrates that stabbing motion exhibits meaningful inter-individual variability and that a model trained on data from a single individual is unlikely to generalise reliably to unseen individuals. From a forensic perspective, this finding underscores the requirement for training datasets that cover a diverse range of participants before any deployment in operational contexts can be considered.

E. Feature Importance

Feature importance scores were extracted from the Random Forest classifiers trained under the random 80/20 split and the LOPO test-P1 condition and are presented in Figure ???. These two conditions were selected to contrast the features exploited when participant data are mixed versus when the model must generalise to an unseen individual.

Under the random split, importance was distributed relatively evenly across the top-ranked features, with orientation standard deviation (*orient_yaw_std*), total acceleration minima (*tacc_x_min*, *tacc_z_min*), and gravity components among the highest contributors. No single feature dominated, suggesting that the classifier exploits a holistic motion signature that covers multiple sensor modalities.

Under the LOPO test-P1 condition, the minimum total acceleration along the vertical axis (*tacc_z_min*) emerged as the single dominant feature, with an importance score approximately double that of the next contributor. This indicates that, when generalising to an unseen individual, the model relies primarily on the downward thrust component of the stabbing motion – the most physically consistent element of the action across individuals – rather than on the broader multi-modal pattern available under within-participant conditions. The prominence of a single feature in the LOPO condition relative to the balanced profile observed under the random split may partially explain the reduced stability of LOPO performance.

Both conditions ranked total acceleration and orientation features among the most informative, consistent with the physical characteristics of the stabbing motion: a rapid directional thrust produces a distinctive total acceleration spike accompanied by wrist rotation, differentiating it from the slower, more oscillatory profiles of the daily activities included in this study.

IV. DISCUSSION

This study examined whether smartwatch motion sensor data can distinguish stabbing motions from visually and kinematically similar everyday activities, with particular attention to forensic relevance. Across all three evaluated classifiers,

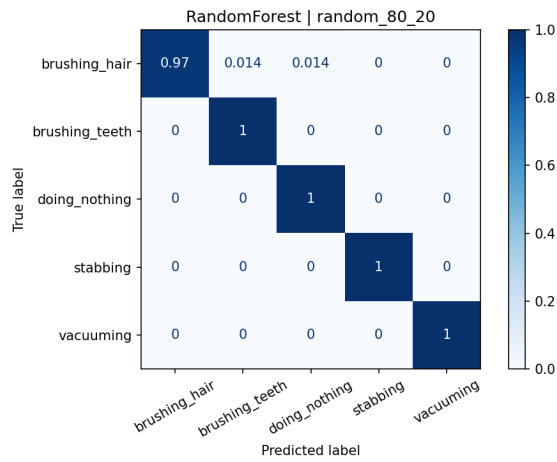


Fig. 1. Normalised confusion matrix — Random Forest, random 80/20 split. Rows represent true activity classes; columns represent predicted classes. Cell values indicate the proportion of true instances classified into each predicted class.

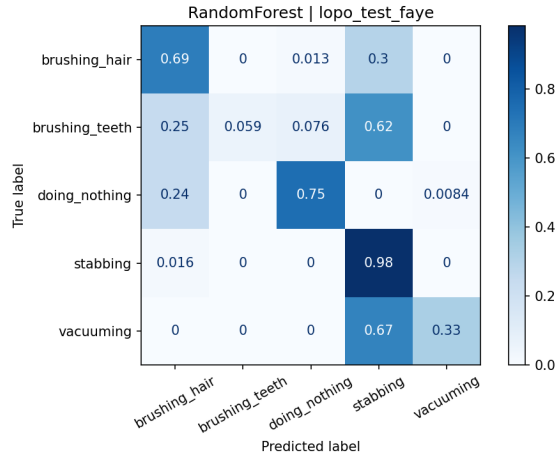


Fig. 2. Normalised confusion matrix — Random Forest, LOPO (train P2, test P1).

stabbing proved to be the most distinctive activity class, though performance depended strongly on the evaluation strategy and the diversity of the training data.

Under the random 80/20 split, all models achieved near-perfect performance, with overall accuracy above 99% and stabbing recall of 1.000. These results indicate that the extracted feature set contains a strong and learnable signal when training and test data originate from the same participant population. However, because sliding windows were generated with 50% overlap, consecutive windows share half of their samples, and when a random split is applied, overlapping windows may appear in both sets. This can lead to optimistic performance estimates, and the random split results should therefore be interpreted as an upper bound rather than an operational estimate.

The LOPO evaluation provides a more conservative and

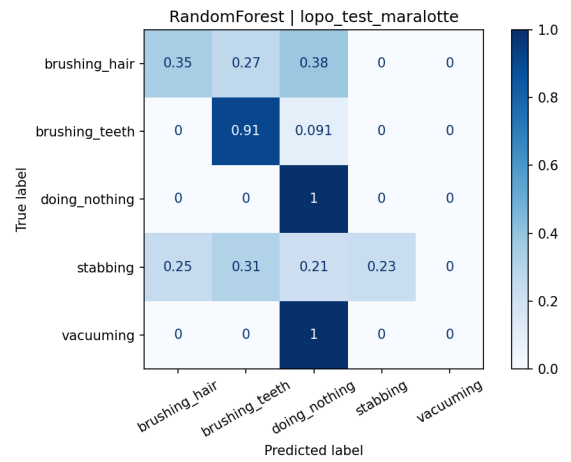


Fig. 3. Normalised confusion matrix — Random Forest, LOPO (train P1, test P2).

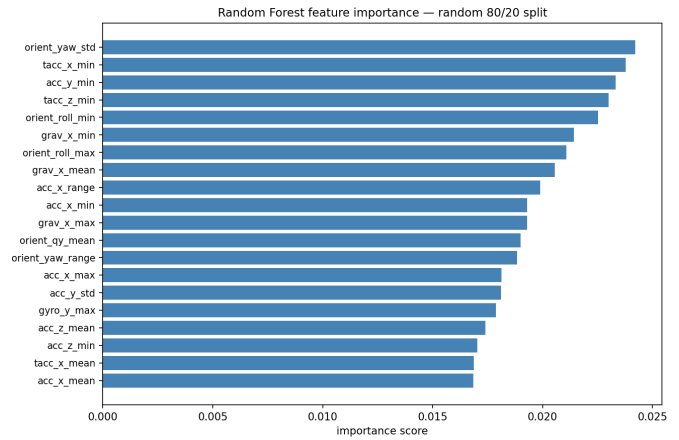


Fig. 4. Top 20 feature importances — Random Forest, random 80/20 split. Importance expressed as mean decrease in impurity, normalised across all features.

forensically meaningful assessment. When models were trained on Participant 2 and tested on Participant 1, overall accuracy decreased substantially but stabbing recall remained high at 0.984, indicating that the stabbing motion produces a sufficiently stable sensor signature to transfer to an unseen individual when the training data includes recordings from more than one person. When the direction was reversed, stabbing recall dropped sharply to between 0.107 and 0.232. Precision remained very high, meaning that when the model predicted stabbing it was usually correct, but the majority of true stabbing windows were not detected. This asymmetry indicates that stabbing motion varies between individuals, and that training data from a single individual does not provide sufficient coverage for reliable cross-person detection. From a forensic perspective, this suggests that a practical system would require training data from a larger and more diverse group of participants before reliable conclusions could be drawn.

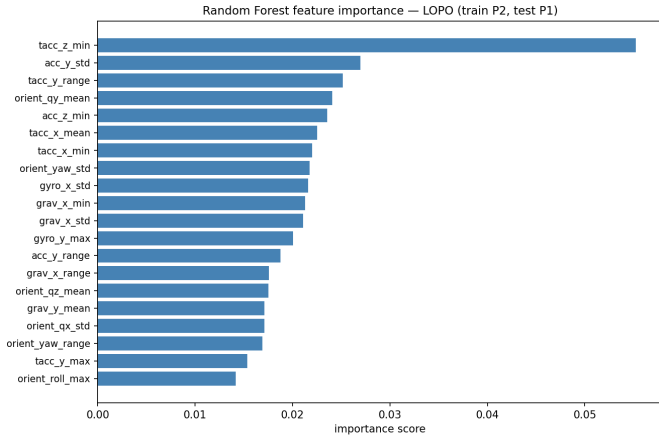


Fig. 5. Top 20 feature importances — Random Forest, LOPO (train P2, test P1).

Feature importance analysis supports this interpretation. When participant data were mixed, the models relied on a combination of orientation, gravity, and total acceleration features. When generalising to an unseen participant, the models relied more heavily on a smaller set of features, with the minimum total acceleration along the vertical axis dominating. This suggests the classifier depended primarily on the most physically consistent component of the stabbing motion – a rapid, directed thrust accompanied by wrist rotation – which is consistent with its expected biomechanics.

Several limitations must be considered. The most important is the small participant pool: with only two effective participant groups, the LOPO results rely on a single train–test pairing in each direction, which is insufficient to draw general conclusions about population-level generalisation. Recording reliability represents a further limitation: intermittent gaps caused by OS-level process suspension and Bluetooth connectivity issues resulted in the loss of entire sessions, contributing to uneven class coverage and imbalances across activity classes. The vacuuming class in particular was severely under-represented with 79 windows, all recorded under elevated heart rate conditions, and showed consistently poor performance in LOPO evaluation.

Heart rate data were excluded after it was found that obtaining high-resolution recordings required running the Samsung Health exercise mode simultaneously with SensorLogger, which introduced additional gaps beyond those caused by the OS alone. As a result, the present study does not evaluate the effect of physiological state on motion classification.

The class imbalance in the random 80/20 split, combined with the window overlap issue, means that the near-perfect random split results should be interpreted with care. Furthermore, all recordings were performed in controlled conditions. The stabbing motion was simulated against an office chair, and real situations may involve resistance from another person, emotional stress, fatigue, grip variation, or differences in clothing, all of which could alter the sensor signature and reduce classification performance in practice.

Variations in how the watch is worn represent an additional source of variability not controlled for in this study. Differences in strap tightness and wrist position can influence gyroscope and orientation measurements [24], introducing inter-individual differences that may affect model performance. Future work should either standardise watch placement or explicitly model this variability.

Forensic applicability is also limited by the data acquisition method. The raw sensor streams used in this study were obtained via SensorLogger, which would not normally be present on a suspect’s device in real casework. Standard forensic acquisition methods typically retrieve data stored by manufacturer applications, and access to high-resolution inertial sensor streams may not be available through these channels. The present study should therefore be regarded as a proof-of-concept demonstrating technical feasibility under research conditions rather than a workflow ready for operational forensic deployment.

Despite these limitations, the results suggest that smartwatch sensor data may have value as supportive evidence in forensic event reconstruction. In particular, orientation and gravity measurements describe the position of the wrist relative to the gravitational field and could potentially complement other forensic methods. For example, in future work, such data could be combined with bloodstain pattern analysis to support or refute proposed scenarios of how a stabbing event occurred. There is also potential for this type of analysis to contribute to investigative work more broadly, though such applications would be computationally and analytically intensive, and would require substantially larger and more diverse datasets before results could be considered reliable.

Finally, ethical considerations must be acknowledged. All data in this study were collected from consenting researcher-participants in a controlled environment, and no real criminal case data were used. Any future operational use of wearable sensor data would require careful legal and ethical evaluation, including appropriate consent frameworks, validation on large and representative datasets, and safeguards against misuse such as inappropriate surveillance or over-interpretation of motion data.

V. CONCLUSION

This study demonstrates that motion sensor data recorded from a smartwatch can be used to distinguish stabbing motions from everyday activities under controlled conditions. Across three tree-based ensemble classifiers, stabbing produced the most consistent and detectable signal in the dataset. While near-perfect performance was obtained under a random train–test split, the more forensically meaningful leave-one-participant-out evaluation showed that generalisation to unseen individuals remains challenging.

The most important finding is the asymmetry observed between the two LOPO conditions. Models trained on data from multiple individuals detected stabbing in an unseen participant with high recall, whereas models trained on a single individual failed to generalise reliably to others. This

indicates that stabbing motion contains both consistent and person-specific components, and that diverse training data are necessary for robust detection. For forensic applications, this suggests that reliable deployment would require datasets from many participants under varied conditions.

Several limitations restrict the strength of the conclusions, including the small participant pool, recording gaps caused by OS-level suspension and Bluetooth connectivity issues, class imbalance, the use of a third-party logging application not typically available in casework, and the use of simulated motions in a controlled environment. These constraints mean that the present work should be interpreted as a proof-of-concept demonstrating technical feasibility rather than evidence of immediate operational applicability. It is due to constraints in time and resources that more activities were not included in the study, and further activities should be assessed before any move can be made toward employing such methods in a forensic context.

Future research should include larger participant groups, improved recording reliability, and validation under more realistic conditions. Further work should also investigate whether manufacturer-stored data obtainable through forensic extraction tools or subject access requests can provide sufficient sensor resolution for similar analyses. Combining motion data with other forensic techniques, such as bloodstain pattern analysis, may further improve the reconstruction of events.

In summary, smartwatch inertial sensor data shows clear potential as an additional source of digital trace evidence, but substantial further validation is required before such methods could be considered reliable for use in operational forensic contexts.

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